Approval Sheet

Product Specification

Part Description: **Bluetooth Module**
Customer Part No:
MOVON Part No: MD-BNC400

Acknowledgement of reception

We have received the attached specification

Date 2020. 08 .04

Company

MOVON Corporation.

Dept.

MOVON R&D LAB

Representative

Mason.Yoo

Received by

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Revision Record

No	Date	Item	Modifications	Approved
1	2020. 08. 04		First Issued	

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1. Module Features

- Low power Bluetooth low energy technology v5.1 specification compliant radio.
- ARM cortex – M4 32-bit processor with FPU, 64Mhz
- Built-in Reference Clock: 32MHz
- RoHS Compliant

2. Applications

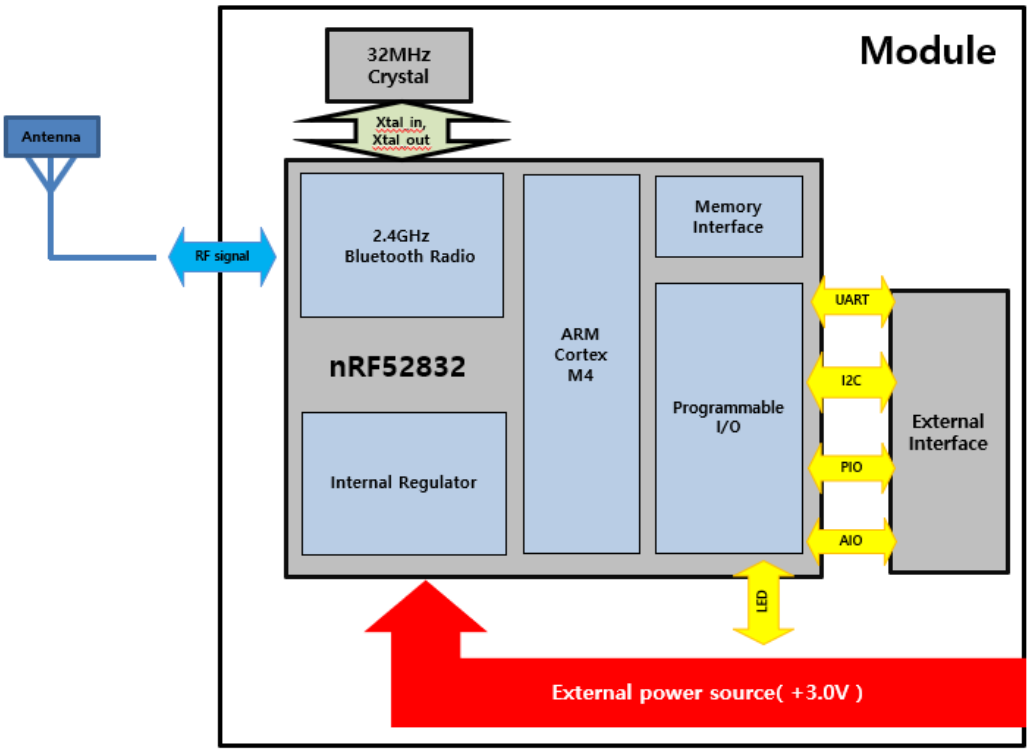
- Internet of Things (IoT) : Home automation, Sensor networks
- Personal area networks : Health/fitness sensor, medical devices
- Interactive entertainment devices : Remote controls, Gaming controllers
- Computer peripherals and I/O devices : Mouse, Keyboard, Gaming

3. Features

- Size (13mm X 12mm X 3mm)
- Surface Mountable

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Overview



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4. Electrical Characteristics

Absolute Maximum Ratings			
Parameter	**Min**	**Max**	**Unit**
Storage Temperate	-40	+125	℃
VDD_BAT	0	3.9	V
VDD_PADS I/0 supply voltage	0	3.9	V

Recommended Operating Conditions			
Parameter	**Min**	**Max**	**Unit**
Temperature	-40	+85	℃
VDD_BAT	3.0		V
VDD_PADS	3.0		V

5. Power Consumption

Parameter	Test Conditions	Min	Typ	Max	Unit
Current Consumption	Advertising(Paring) Mode	-	TBD	-	mA
	Sleep Mode (Connection)	-	TBD	-	μA
	UART active & Transmit data	-	TBD	-	mA
	Power On (No Connection)	-	TBD	-	μA

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6. Pin Description

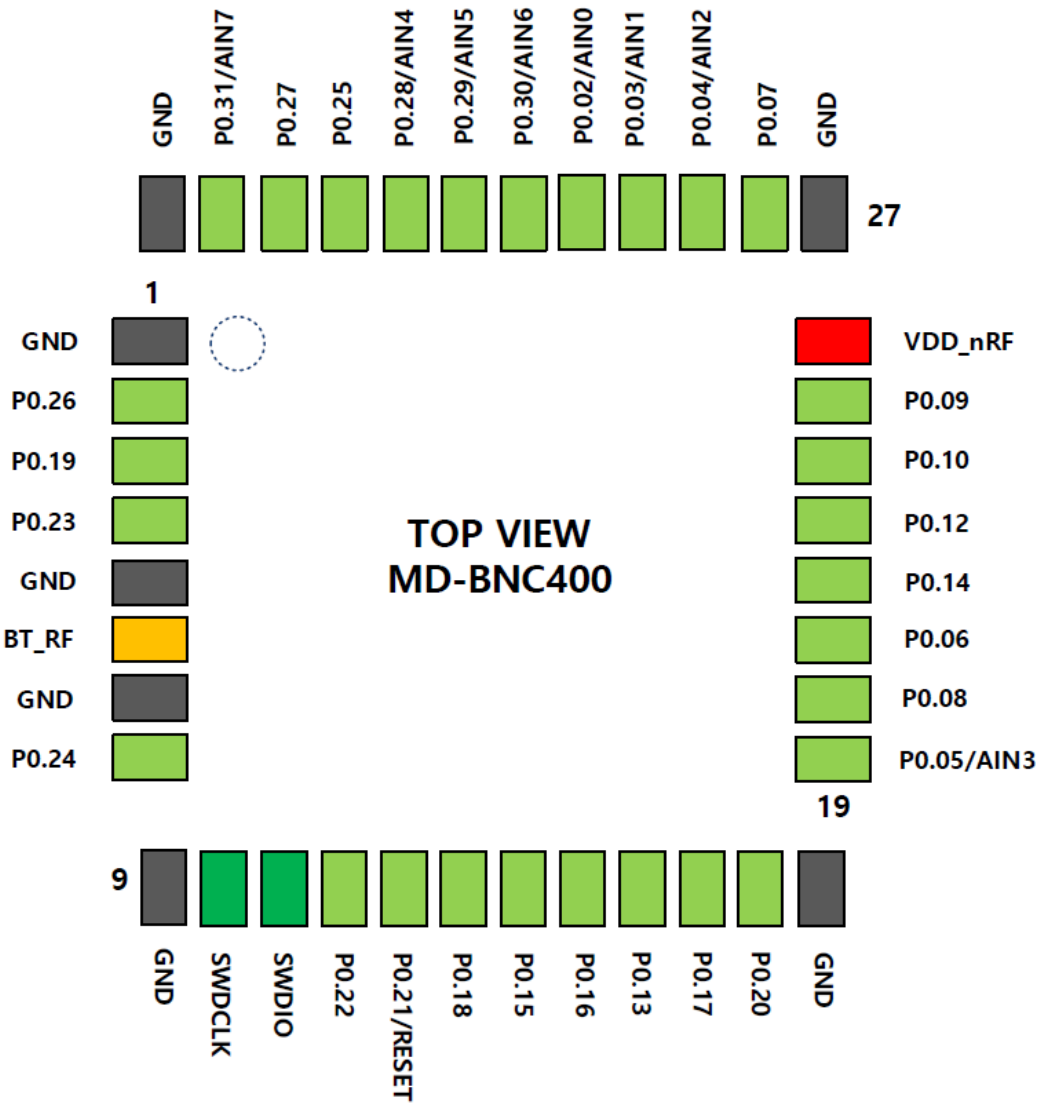
PIN No.	Pin Name	Description	Pad Type
1	GND	Ground connections	
2	P0.26	General purpose I/O	
3	P0.19	General purpose I/O	
4	P0.23	General purpose I/O	
5	GND	Ground connections	
6	BT_RF	Bluetooth 50Ω transmitter output / receiver input	RF
7	GND	Ground connections	
8	P0.24	General purpose I/O	
9	GND	Ground connections	
10	SWDCLK		
11	SWDIO		
12	P0.22	General purpose I/O	
13	P0.21/RESET	General purpose I/O Configurable as pin reset	
14	P0.18	General purpose I/O	
15	P0.15	General purpose I/O	
16	P0.16	General purpose I/O	
17	P0.13	General purpose I/O	
18	P0.17	General purpose I/O	
19	P0.20	General purpose I/O	
20	GND	Ground connections	
21	P0.05/AIN3	General purpose I/O	
22	P0.08	General purpose I/O	
23	P0.06	General purpose I/O	
24	P0.14	General purpose I/O	
25	P0.12	General purpose I/O	
26	P0.10	General purpose I/O	
27	P0.09	General purpose I/O	
28	VDD_NRF		
29	GND	Ground connections	
30	P0.07	General purpose I/O	

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31	P0.04/AIN2	General purpose I/O	
32	P0.03/AIN1	General purpose I/O	
33	P0.02/AIN0	General purpose I/O	
34	P0.30/AIN6	General purpose I/O	
35	P0.29/AIN5	General purpose I/O	
36	P0.28/AIN4	General purpose I/O	
37	P0.25	General purpose I/O	
38	P0.27	General purpose I/O	
39	P0.31/AIN7	General purpose I/O	
40	GND	Ground connections	

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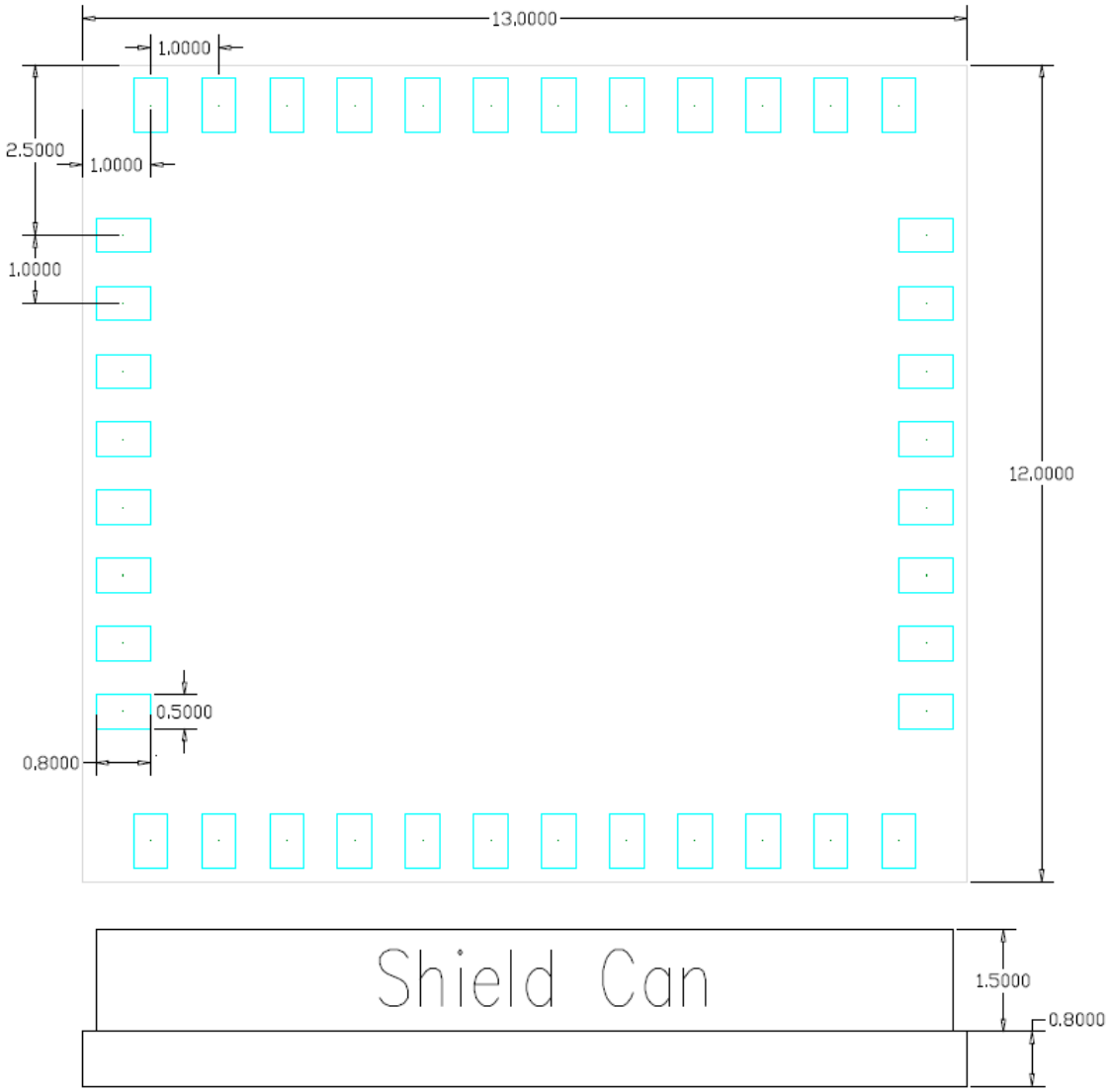
7.PIN MAP



<Top View>

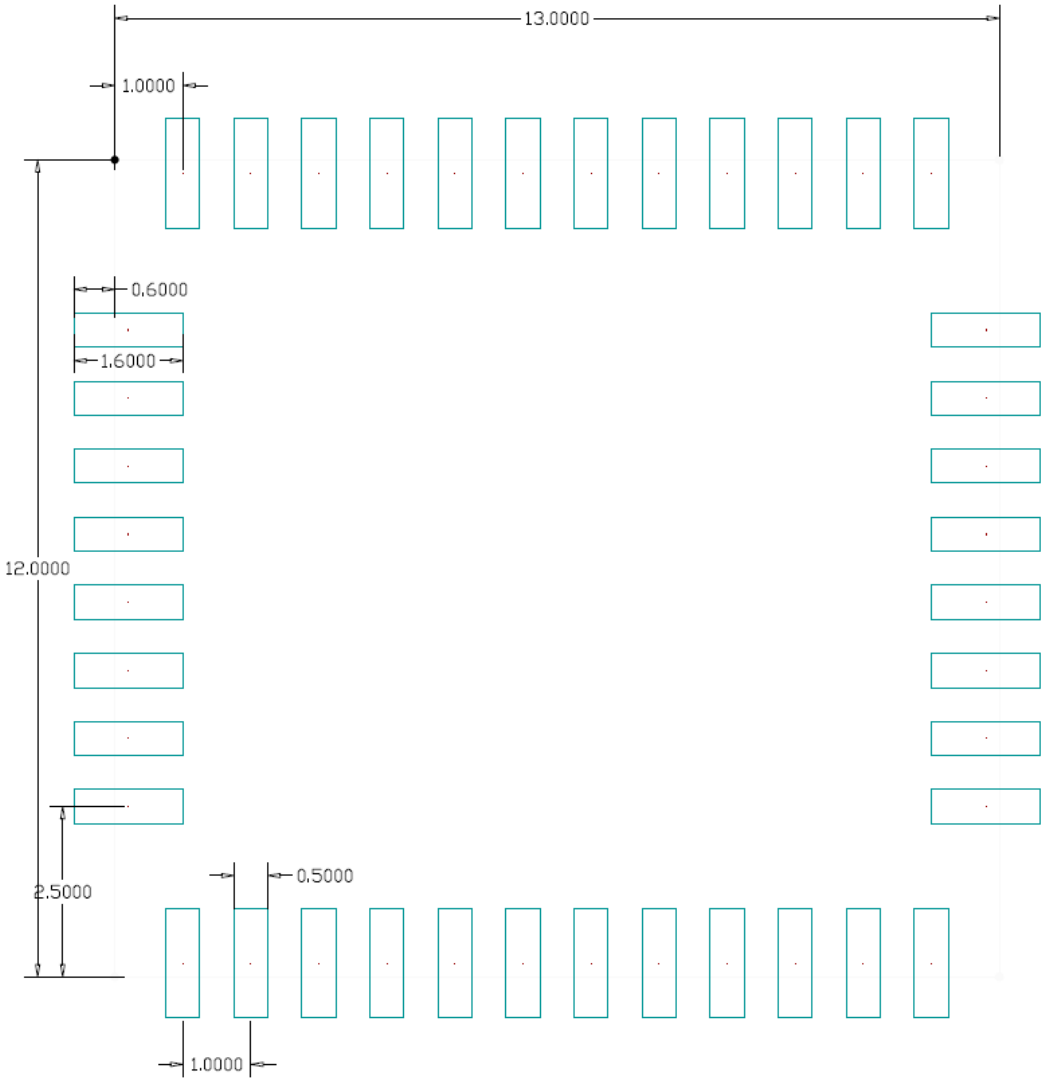
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8. Dimension(unit :mm)



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9. Recommended PCB Pattern(Unit : mm)



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